Convolutional Neural Network to Classify Computer Generated vs. Real Human Faces

Andre Barle
Yifei Xue

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1 Introduction

The rise of artificial intelligence (AI) has led to a significant challenge: distinguishing real images from those generated by artificial intelligence. With the evolution of computation power and image generation models, such as Generative Adversarial Networks (GAN), it becomes increasingly difficult to distinguish between genuine and synthetic content. Multiple fields could potentially benefit from better identifying AI images, including media authenticity, law enforcement with fraud detection, and improving artificial image generation.

Numerous studies have been done with similar topics, though mostly focused on video data rather than still images. In our study, we aim to address this issue by training a deep learning model to classify whether an image of a person's face is real or AI generated. Specifically, we utilized a robust multilayer convolutional neural network (CNN) to deal with the binary classification issue between real and fake human faces.However, due to the nature of neural networks, explaining the key differences between the two classes is nearly impossible. To resolve this issue, we introduced gradient-weighted class activation mapping (Grad-CAM) technique to interpret the difference between two classes. This approach provides more explanations and greater transparency in the classification process.

1.1 Literature Review

In terms of feasibility, previous studies done in the field focus on artifact detection via keypoint analysis. These artifacts commonly appear on the teeth and eyes, where they are blurred and inconsistent with human features. Another technique is called Gradient Class Activation Mapping. This technique can provide a heatmap over a given image, showing areas of focus for the neural network. Another approach which yields strong results is pairwise learning fake vs real images. By combining two neural networks, we can narrow down the area of focus in the image based on where the previous network has learned to look for artifacts. This can increase the speed and precision of training for the final classifier neural network. The literature also suggests the usage of multiple data sources generated from a variety of AI generative networks to reduce overfitting. We plan to test and employ these techniques in order to increase the precision of the final model.

1.2 Hypothesis

New fake images which can deceive a human observer still hold predictable patterns in terms of artifacts and defects that deep learning can help uncover. Defects on new fake faces have become more smoothed out and require a broader attention zone for the computer to determine whether the image is real or fake.

2 Methods

In this project, pairwise learning did not yield meaningful results because we did not have exact image pairs to compare. For this method to work properly, it would be ideal to have a real image, and then the deepfake version of the image, however this did not align with our datasets and with the ultimate goals of this project. Before the actual training of the CNN model, multiple data preprocessing steps were taken. The data was aggregated from four distinct datasets to ensure varieties and was further split into training, testing, and validation sets. Each dataset was diverse in terms of the types of deepfake/fake images and real images of people it contained. Binary labels were used to identify the class, with 1 indicating the image was real and 0 indicating it was AI-generated or fake. All images were resized to (128, 128) for consistency. We also normalized the pixel values to the range [0, 1] by dividing by 255. By combining real and fake images, we ensured a balanced class distribution in the dataset.

We are interested in understanding the evolution of Generative Adversarial Networks (GAN) in generating fake images of human faces. To aid this task, we trained two separate convolutional neural networks using four datasets. The first convolutional neural network 'model without fake' was trained on data which excludes any images of the 'All These People Dont Exist' dataset, and was then asked to predict on a test set which includes examples of real faces as well as fake faces from the 'All These People Dont Exist' dataset. The second convolutional neural network 'model without fake' was trained on data which included fake images from the 'All These People Dont Exist' dataset in the training and validation phases.

We used the tensorflow.keras library to train both neural networks. Both convolutional neural networks were prepared for training using ImageDataGenerator for fast image processing. Additionally, both networks were trained with three convolutional layers, each followed by max-pooling layers with a size of (2, 2). The first convolutional layer had 32 filters, the second had 64, and the third had 128. A consistent filter size of (3, 3) with the Rectified Linear Unit(ReLU) activation function was applied in each convolutional layer. They were optimized in the same way with Adam optimizer and used binary cross-entropy as the loss function, over three epochs. The architecture of both neural networks was the same, the only difference was in the training approach, where in the 'model without fake', we use the 'All These People Dont Exist' dataset only in the test dataframe, and do not feed any image example of this dataset to the network in the training process.

By comparing and contrasting the activations of each convolutional neural network using Gradient Class Activation Mapping, we are able to discern the differences in terms of learned attention zones between a neural network trained on more primitive fake images vs a neural network trained with examples of more sophisticated fake images.

2.1 Dataset Description

Parts of dataset 1 and 2 will be used to assess final performance. The final goal is for the model to classify every image in dataset 2 as fake. We plan to merge and use datasets 3 and 5 as training data for feature training the first neural network and supplement with dataset 1, 2, and 4 as needed to handle potential class imbalance.

- Dataset 1 - 52k images of real people
- Dataset 2 - 6,873 images of fake people
- Dataset 3 - 70k real and 70k fake images for training
- Dataset 4 - 202,599 real images
- Dataset 5 - 50k real and 50k fake images for training

All datasets have a free license to utilize for research and have been rated well by other users, some have been referenced in the literature as well.

3 Experiments

By comparing the models using Gradient Class Activation Mapping, we can observe the specific zones of attention that help the neural network to make a decision regarding classifying an image as real vs fake. (Figure 2, Figure 3) Zones that are lit up with bright colors are zones with more value to the neural network in determining if an image is real or fake. On the left column are real images with the attention mapping of the real image shown next to it. On the right column are fake images with the attention mapping of the fake image shown next to it. From this, we observe some differences in how the neural networks are classifying real images vs fake images and we can tell that there may be some difference in how each network is operating on determining if an image is real or fake.

In terms of metrics, the model trained without the 'All These People Dont Exist' Dataset scored on Accuracy: 0.929, Precision: 0.937, F1-score: 0.947, and

Recall: 0.958. The other model trained on this dataset scored Accuracy: 0.952, Precision: 0.960, F1-score: 0.966, Recall: 0.972. These results correspond to the confusion matrices presented in Figure 7 and Figure 8. The second model performed better, as expected since it was trained on samples of the 'All These People Dont Exist' Dataset, giving it a more diverse sampling of fake images. More important than the comparison in metrics is the comparison in learning from each model. The more advanced model picks up on other features other than the eyes and mouth area in order to classify images. This is a signal that new fake images have corrected some of the visual distortions typically found in specific regions around the eyes and mouth, and require a broader idea of the face in order to correctly classify.

To aid this visualization of difference, we can sample 5000 images from the test set for both neural networks and render an average heatmap of which zones the neural networks find more valuable in determining if an image is real or fake. (Figure 4, Figure 5) We find only subtle differences in how the model trained without the 'All These People Dont Exist' dataset treats images that are real vs fake. Most of the attention is focused on a specific point around the eyes as well as the general area around the mouth. This highlights the fact that older fake images often had visible defects rendered in the eye region with some defects present around the mouth. It seems these small defects/artifacts were a key factor in allowing the more primitive model to classify images.

Interestingly, in Figure 5 we observe a marked difference in how the more advanced neural network determines if an image is real or fake. More regions around the face and neck area are activated in general. For real images, the neck, forehead, nose, and mouth are more important to the advanced model when compared to the more primitive version. In classifying a fake image, more face regions are activated in the advanced model, implying that many of the defects/artifacts present in older fake images on the eyes have been improved upon. The model is requiring more information about the general attributes of the face before it is able to make an informed decision. However, there are still very persistent hot spots around the eyes and mouth which are important to classification.

Finally, we can observe on some select images how one neural network classifies vs the other. Figure 6 shows the attention zones of 'model without fake' in red and 'model with fake' in green. Grey regions are those where both models are similarly activated. In this visualization, we can observe some differences in where each model finds its predictive power. The model trained on the additional advanced fake images has a more general focus on the contours of the face, and includes more activation on the forehead and less on the eyes and mouth. This implies that fake images have improved in quality overall in regions where they used to produce reliable defects/artifacts. A more well rounded understanding of the face is required in order for a model to make a classification decision.

4 Conclusion

In final analysis, we have demonstrated our hypothesis is true by comparing the Gradient Class Activation Mapping between two neural networks. By providing new fake images, we trained the neural network to recognize more complex facial features when it is performing classification. This comparison gives us a glimpse into where fake face generation has improved over the years and where it is still lacking in smoothing out imperfections. In Figure 5 we notice there are still problems in generating eyes and mouth features more than there are problems in generating foreheads and other smooth features. GAN generated images still have deformities wherever there are edges on a face.

We have also shown that a neural network that is only trained on old fake images can still effectively classify new fake images that are much harder to discern with the human eye. This means that at the pixel level there are still patterns which the computer can notice but we are unable to notice ourselves. Thus, in the future as GAN generated fake images improve, we will increasingly need to rely on neural networks to classify the images as real or fake. This research suggests that we will always be able to create a neural network which can classify fake images vs real ones, as long as we continue to train models on the latest fake images. Even with a model trained on old data, we have a greater chance to discern these new fake images that with the human eye alone.

4.1 Future work

For future research, it would be interesting to use a face detection algorithm to help create an attention mask for the face in an image. This could further improve the neural network and yield more precise results in the Gradient Class Activation Mapping. Additionally, the usage of superior hardware and more data would improve the results of this research. We can also use this research to measure the rate of progress in fake face generation. By measuring the difference in Gradient Class Activation Mapping, we can better understand where fake image GANs are improving and where they are still struggling. Finally, future researchers can use this project to help identify fake images for purposes of maintaining ethics and safety in social media platforms. Conversely, this can also be used to help GANs improve in certain areas of fake image generation. This Activation mapping can be used to reinforce a GAN model to spend more time on rendering regions that are typically prone to defects/artifacts.

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5 Charts and Figures

Figure 1: Real image (left), Fake image(right)

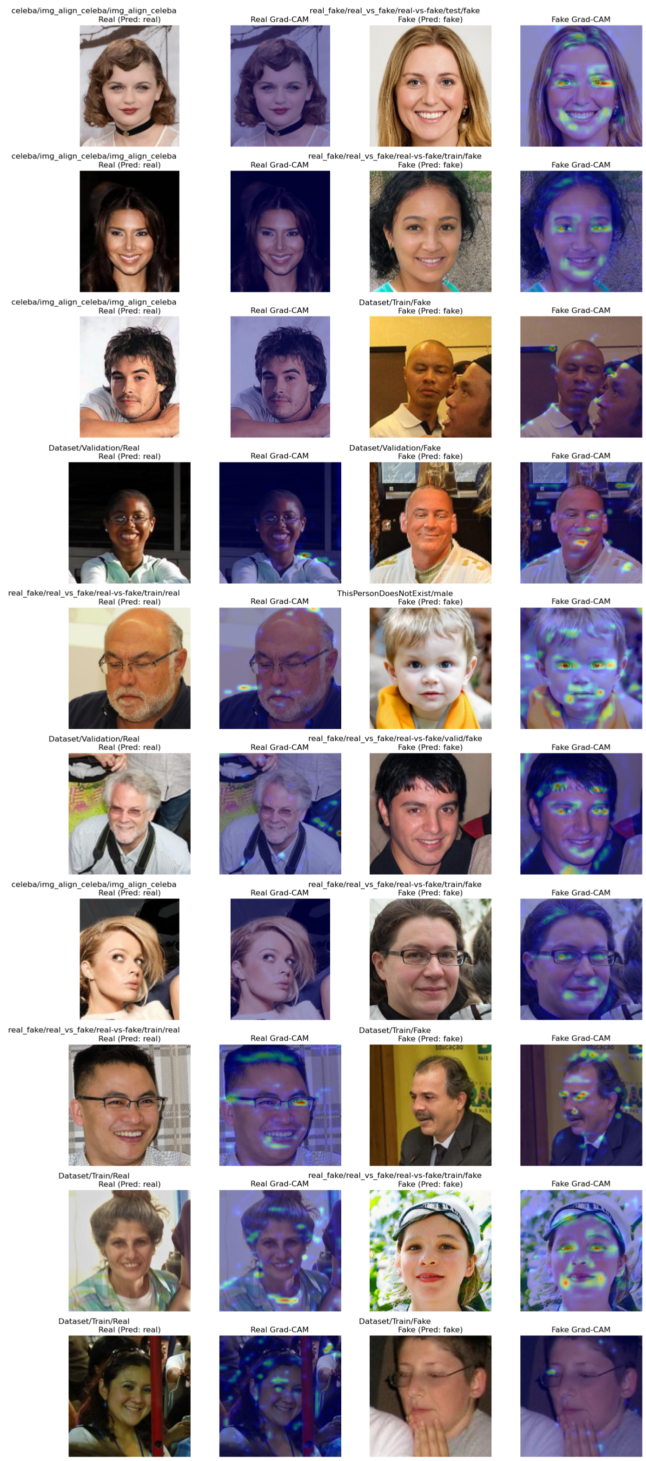

Figure 2: Gradient Class Activation Mapping for model trained without Face Dataset Of People That Don't Exist. Real image (left column) with Gradient Class Activation Mapping in dark blue to the right of the image, Fake image(right column) with Gradient Class Activation Mapping in dark blue to the right of the image

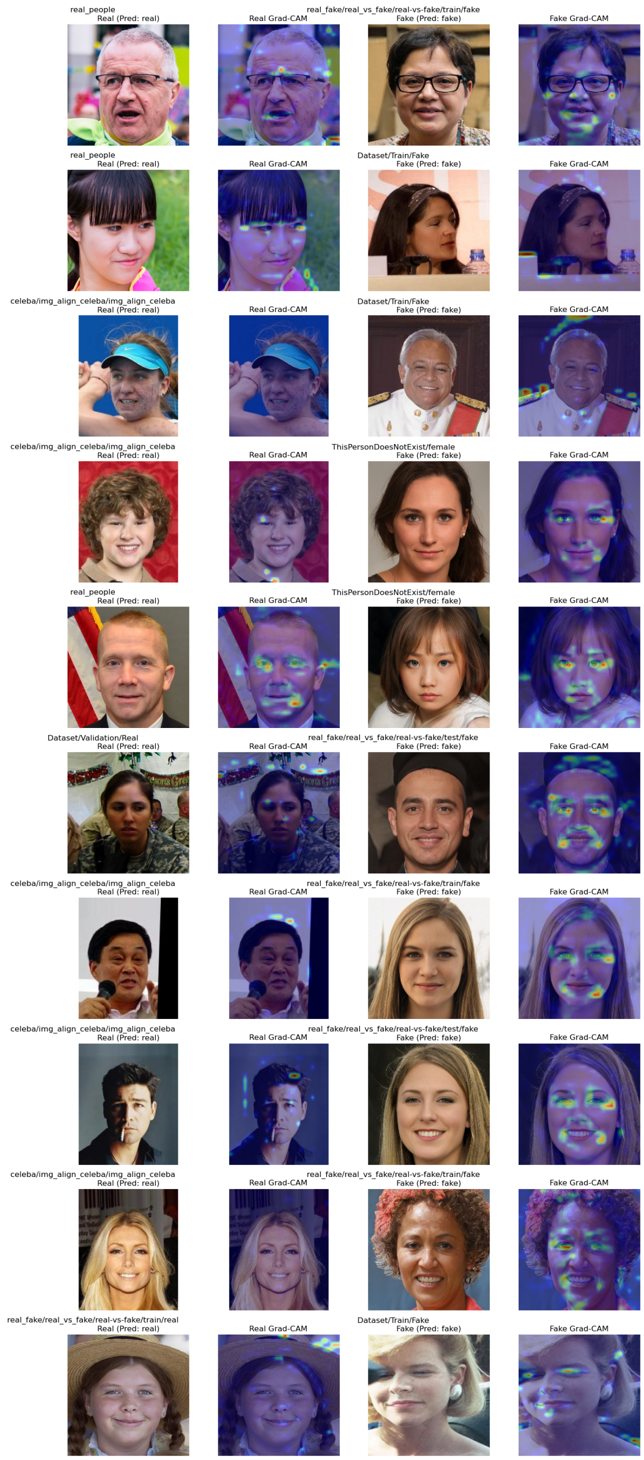

Figure 3: Gradient Class Activation Mapping for model trained with Face Dataset Of People That Don't Exist. Real image (left column) with Gradient Class Activation Mapping in dark blue to the right of the image, Fake image(right column) with Gradient Class Activation Mapping in dark blue to the right of the image

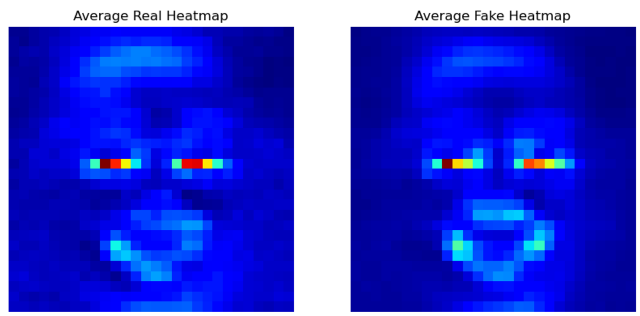

Figure 4: Average Gradient Class Activation Mapping for model trained without Face Dataset Of People That Don't Exist. Sampled 5000 times

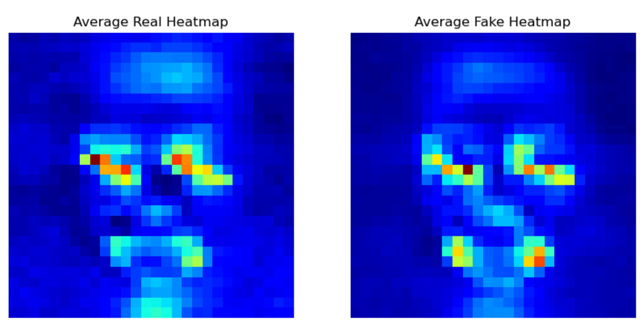

Figure 5: Average Gradient Class Activation Mapping for model trained with Face Dataset Of People That Don't Exist. Sampled 5000 times

Figure 6: Difference between models in Gradient Class Activation; red - model not trained on Face Dataset Of People That Don't Exist, green - model trained on Face Dataset Of People That Don't Exist fake, grey - similar in both models

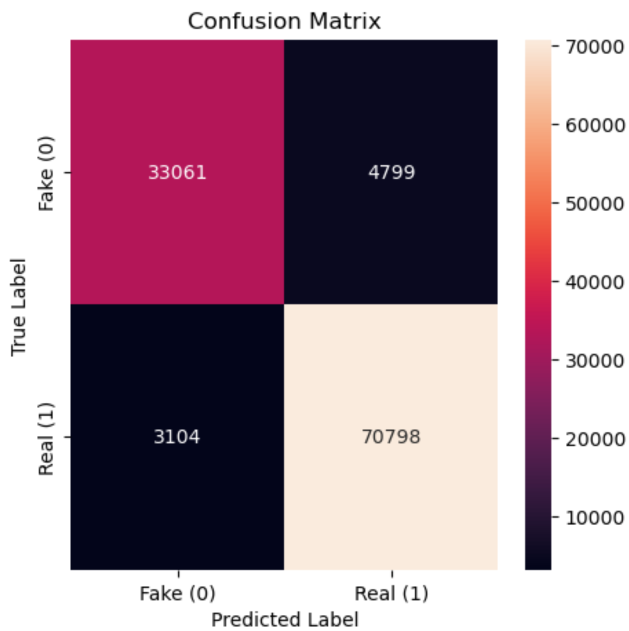

Figure 7: Confusion matrix of model trained without Face Dataset Of People That Don't Exist

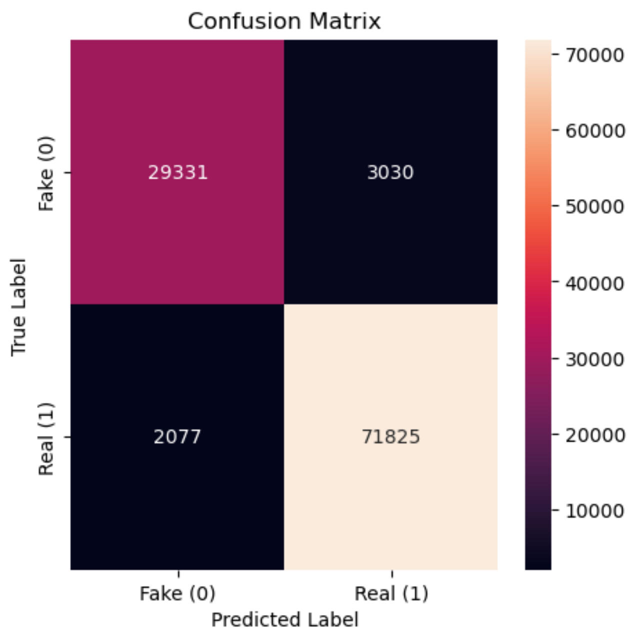

Figure 8: Confusion matrix of model trained with Face Dataset Of People That Don't Exist